

Monday Warm Up: Unit 5, Electromagnetism and Optics

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1 Memory Bank

- **Optics:** Snell's Law states that, for an interface between two media with *indices of refraction* n_1 and n_2 , light leaves one medium with an angle θ_1 and enters the second medium with an angle θ_2 as follows:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (1)$$

The two angles are defined with respect to vertical.

- **Optics:** Fresnel's equations give the fraction of power (in light) that is reflected (R) or transmitted (T) by an interface between media with indices of refraction n_1 and n_2 :

$$R = \left| \frac{n_1 - n_2}{n_1 + n_2} \right|^2 \quad (2)$$

Note that, to conserve power, $T = 1 - R$.

- **Thin lens equations.** Let an object be a distance d_o from the center of a thin lens. If the lens produces a real image at a distance d_i from the center of the lens on the opposite side from the object, d_o and d_i are related by the *focal length* f :

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} \quad (3)$$

If the object has a characteristic size h_o and the image has a characteristic size h_i , the *magnification*, m , is

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o} \quad (4)$$

- **Total internal reflection** occurs when light no longer refracts out of a material. This occurs when the angle of incidence is greater than or equal to

$$\theta_C = \sin^{-1}(n_2/n_1) \quad (5)$$

2 Fiber Optics, Snell's Law, and Total Internal Reflection

1. A light ray entering an optical fiber surrounded by air is first refracted and then reflected as shown in Fig. 1. Show that if the fiber has $n_2 = 1.5$, with air having $n_1 = 1.0$, any incident ray will be totally internally reflected. *Hint: pick a reasonable number for θ_1 , and work out θ_2 and θ_3 .*

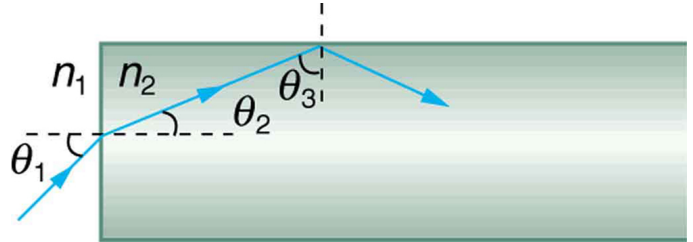


Figure 1: A light ray enters a fiber optic cable.

3 Lens Optics

1. A certain slide projector has a 100 mm focal length lens. (a) How far away is the screen, if a slide is placed 103 mm from the lens and produces a sharp image? (b) If the slide is 24.0 by 36.0 mm, what are the dimensions of the image?
2. A dentist examines a patient using a 15.0 cm focal length eyepiece held 13.5 cm from their tooth. (a) Where is the image? (b) What is its magnification? (c) How big is the image of a 10.00 mm diameter tooth?